

APM1110 Formative Assessment 7

Calalang, Jhan Cazhandra D., Ingaran, Lord Arvin C., Peña, Angelina Jolie C.

2026-10-08

PART 2

APPLYING NORMAL DISTRIBUTION ON CAMPUS

Nursing Students' Midterms Sleep Duration

1. IMPORT THE DATA

```
library(readxl)
sleep_data <- read_excel(
  "Nursing Students' Midterms Sleep Duration Survey (Responses).xlsx"
)
```

```
# View data
sleep_data
```

```
## # A tibble: 58 x 2
##   'Do you agree to participate in this survey?' On average, how many hours of-1
##   <chr> <dbl>
## 1 Yes 4
## 2 Yes 5
## 3 Yes 7
## 4 Yes 3
## 5 Yes 5
## 6 Yes 4
## 7 Yes 5
## 8 Yes 7
## 9 Yes 7
## 10 Yes 4
## # i 48 more rows
## # i abbreviated name:
## # 1: 'On average, how many hours of sleep do you get per night? (Just the number of hours without r
```

```
# Check structure
str(sleep_data)
```

```
## tibble [58 x 2] (S3: tbl_df/tbl/data.frame)
## $ Do you agree to participate in this survey?
## $ On average, how many hours of sleep do you get per night? (Just the number of hours without minutes)

# Extract sleep-duration variable
sleep_hours <- as.numeric(
  sleep_data[[2]]
)

# Remove missing values if there are any
sleep_hours <- na.omit(sleep_hours)

# Number of observations
length(sleep_hours)

## [1] 58
```

2. BASIC SUMMARY STATISTICS

```
summary(sleep_hours)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.000   4.000   5.000   5.155   6.000   9.000
```

```
# Mean
mu <- mean(sleep_hours)

# Sample standard deviation
sigma <- sd(sleep_hours)

# Variance
sleep_variance <- var(sleep_hours)

# Median
sleep_median <- median(sleep_hours)

# Mode
sleep_mode <- as.numeric(
  names(
    sort(
      table(sleep_hours),
      decreasing = TRUE
    )[1]
  )
)

# Minimum
sleep_min <- min(sleep_hours)

# Maximum
```

```
sleep_max <- max(sleep_hours)
```

```
mu
```

```
## [1] 5.155172
```

```
sigma
```

```
## [1] 1.694177
```

```
sleep_variance
```

```
## [1] 2.870236
```

```
sleep_median
```

```
## [1] 5
```

```
sleep_mode
```

```
## [1] 4
```

```
sleep_min
```

```
## [1] 2
```

```
sleep_max
```

```
## [1] 9
```

3. FREQUENCY DISTRIBUTION

```
frequency_table <- table(sleep_hours)
```

```
frequency_table
```

```
## sleep_hours
```

```
##  2  3  4  5  6  7  8  9
```

```
##  3  5 16 11  9  8  5  1
```

```
# Convert frequency table into a data frame
```

```
frequency_distribution <- data.frame(  
  Sleep_Hours = as.numeric(names(frequency_table)),  
  Frequency = as.numeric(frequency_table)  
)
```

```
frequency_distribution
```

##	Sleep_Hours	Frequency
## 1	2	3
## 2	3	5
## 3	4	16
## 4	5	11
## 5	6	9
## 6	7	8
## 7	8	5
## 8	9	1

4. HISTOGRAM OF SLEEP DURATION & ADD NORMAL DISTRIBUTION CURVE

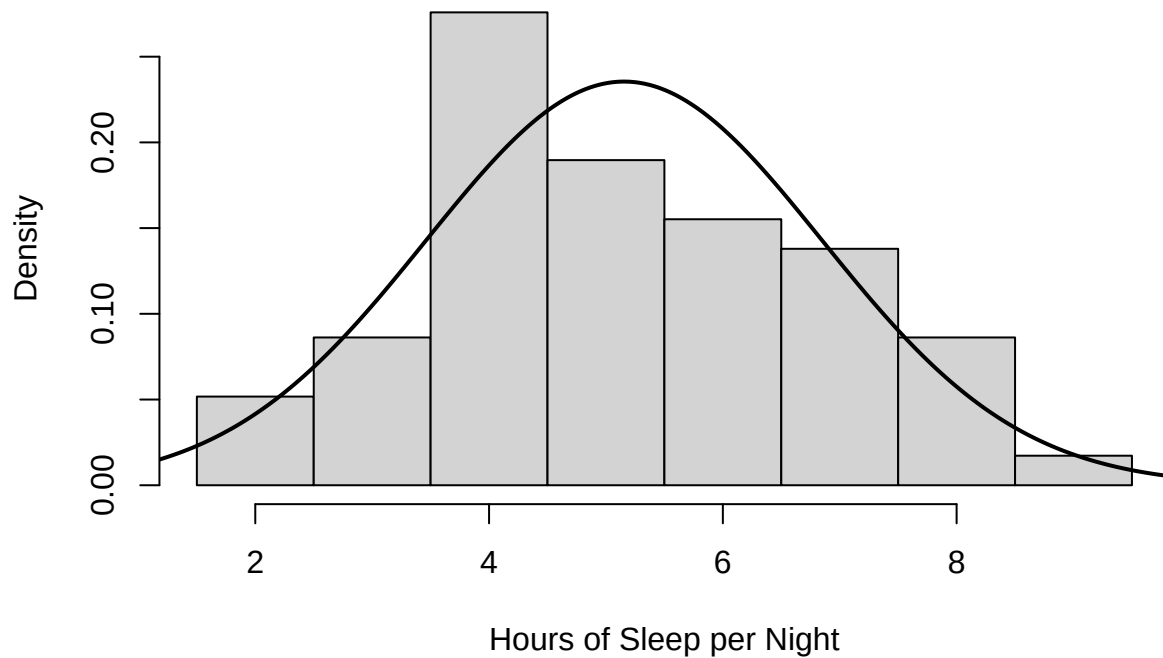
```
hist(
  sleep_hours,
  freq = FALSE,
  main = "Nursing Students' Midterms Sleep Duration",
  xlab = "Hours of Sleep per Night",
  ylab = "Density",
  breaks = seq(
    min(sleep_hours) - 0.5,
    max(sleep_hours) + 0.5,
    by = 1
  )
)

x_normal <- seq(
  min(sleep_hours) - 1,
  max(sleep_hours) + 1,
  length.out = 500
)

normal_curve <- dnorm(
  x_normal,
  mean = mu,
  sd = sigma
)

lines(
  x_normal,
  normal_curve,
  lwd = 2
)
```

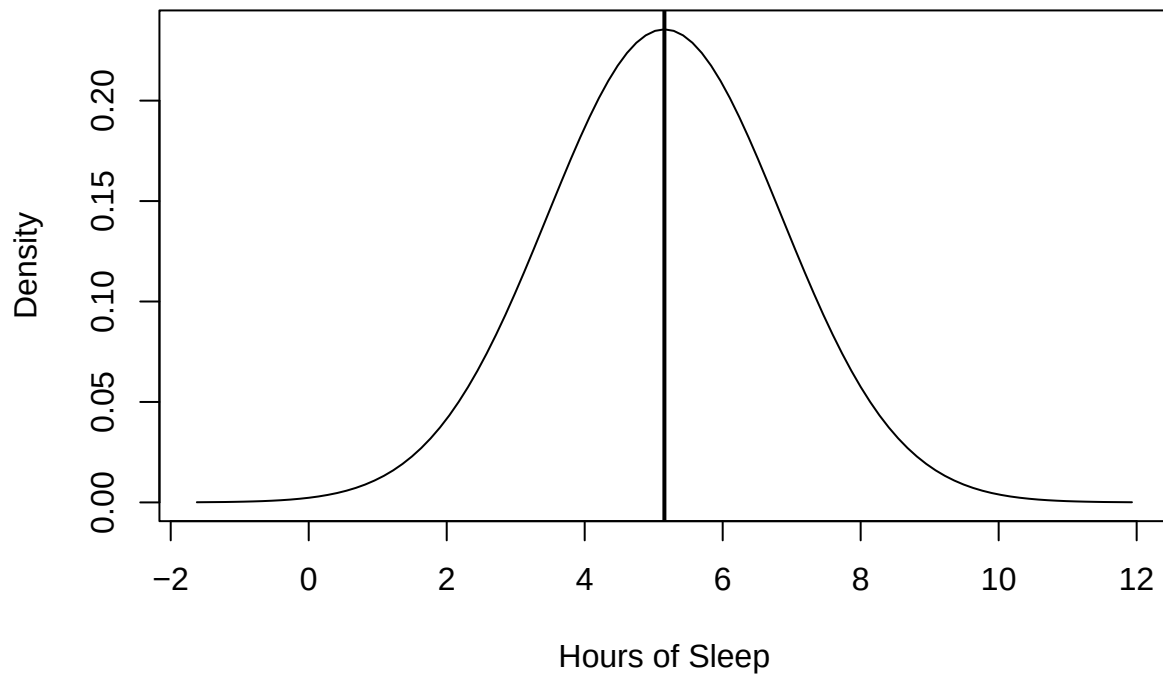
Nursing Students' Midterms Sleep Duration



5. NORMAL DISTRIBUTION CURVE BY ITSELF

```
curve(  
  dnorm(  
    x,  
    mean = mu,  
    sd = sigma  
  ),  
  from = mu - 4 * sigma,  
  to = mu + 4 * sigma,  
  main = "Normal Distribution of Nursing Students' Sleep",  
  xlab = "Hours of Sleep",  
  ylab = "Density"  
)  
  
# Add vertical line at the mean  
abline(  
  v = mu,  
  lwd = 2  
)
```

Normal Distribution of Nursing Students' Sleep



6. CALCULATE 1-SIGMA INTERVAL

```
lower_1sigma <- mu - sigma
upper_1sigma <- mu + sigma

within_1sigma <- sum(
  sleep_hours >= lower_1sigma &
  sleep_hours <= upper_1sigma
)

percent_1sigma <- (
  within_1sigma / length(sleep_hours)
) * 100

lower_1sigma
```

```
## [1] 3.460995
```

```
upper_1sigma
```

```
## [1] 6.849349
```

```
within_1sigma
```

```
## [1] 36
```

```
percent_1sigma
```

```
## [1] 62.06897
```

7. CALCULATE 2-SIGMA INTERVAL

```
lower_2sigma <- mu - 2 * sigma
upper_2sigma <- mu + 2 * sigma

within_2sigma <- sum(
  sleep_hours >= lower_2sigma &
  sleep_hours <= upper_2sigma
)

percent_2sigma <- (
  within_2sigma / length(sleep_hours)
) * 100

lower_2sigma
```

```
## [1] 1.766818
```

```
upper_2sigma
```

```
## [1] 8.543527
```

```
within_2sigma
```

```
## [1] 57
```

```
percent_2sigma
```

```
## [1] 98.27586
```

8. CALCULATE 3-SIGMA INTERVAL

```
lower_3sigma <- mu - 3 * sigma
upper_3sigma <- mu + 3 * sigma

within_3sigma <- sum(
```

```

sleep_hours >= lower_3sigma &
sleep_hours <= upper_3sigma
)

percent_3sigma <- (
  within_3sigma / length(sleep_hours)
) * 100

lower_3sigma

```

```
## [1] 0.07264121
```

```
upper_3sigma
```

```
## [1] 10.2377
```

```
within_3sigma
```

```
## [1] 58
```

```
percent_3sigma
```

```
## [1] 100
```

9. THEORETICAL NORMAL DISTRIBUTION PERCENTAGES

These are the theoretical percentages for a normal distribution.

```

theoretical_1sigma <- pnorm(
  mu + sigma,
  mean = mu,
  sd = sigma
) -
pnorm(
  mu - sigma,
  mean = mu,
  sd = sigma
)

```

```

theoretical_2sigma <- pnorm(
  mu + 2 * sigma,
  mean = mu,
  sd = sigma
) -
pnorm(
  mu - 2 * sigma,
  mean = mu,

```

```

    sd = sigma
  )

  theoretical_3sigma <- pnorm(
    mu + 3 * sigma,
    mean = mu,
    sd = sigma
  ) -
  pnorm(
    mu - 3 * sigma,
    mean = mu,
    sd = sigma
  )

  theoretical_1sigma

```

```
## [1] 0.6826895
```

```
theoretical_2sigma
```

```
## [1] 0.9544997
```

```
theoretical_3sigma
```

```
## [1] 0.9973002
```

10. STANDARDIZE THE DATA

```

z_scores <- (
  sleep_hours - mu
) / sigma

head(z_scores)

```

```
## [1] -0.68184869 -0.09159162  1.08892253 -1.27210576 -0.09159162 -0.68184869
```

11. CHECK FOR OUTLIERS USING IQR

```

Q1 <- quantile(
  sleep_hours,
  0.25
)

Q3 <- quantile(
  sleep_hours,
  0.75
)

```

```

)

IQR_value <- IQR(
  sleep_hours
)

lower_fence <- Q1 - 1.5 * IQR_value
upper_fence <- Q3 + 1.5 * IQR_value

outliers <- sleep_hours[
  sleep_hours < lower_fence |
  sleep_hours > upper_fence
]

Q1

```

```

## 25%
##    4

```

```
Q3
```

```

## 75%
##    6

```

```
IQR_value
```

```
## [1] 2
```

```
lower_fence
```

```

## 25%
##    1

```

```
upper_fence
```

```

## 75%
##    9

```

```
outliers
```

```
## numeric(0)
```

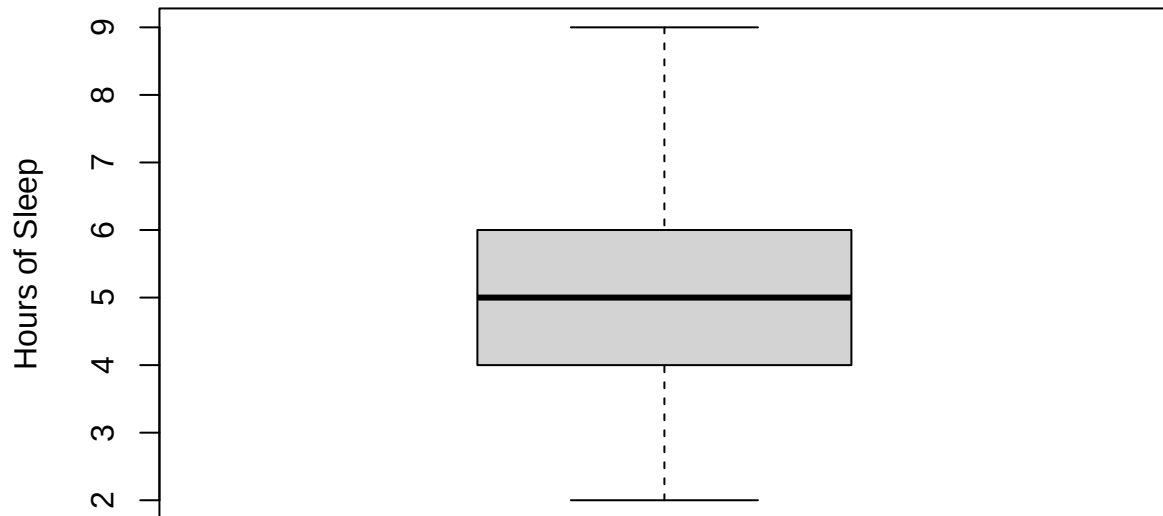
12. BOXPLOT

```

boxplot(
  sleep_hours,
  main = "Boxplot of Nursing Students' Sleep Duration",
  ylab = "Hours of Sleep"
)

```

Boxplot of Nursing Students' Sleep Duration



13. PART 2 SUMMARY TABLE

```
part2_summary <- data.frame(  
  
  Statistic = c(  
    "Number of observations",  
    "Mean",  
    "Median",  
    "Mode",  
    "Sample standard deviation",  
    "Variance",  
    "Minimum",  
    "Maximum",  
    "Within 1 SD",  
    "Within 2 SD",  
    "Within 3 SD"  
  ),  
  
  Value = c(  
    length(sleep_hours),  
    mu,  
    sleep_median,  
    sleep_mode,
```

```

    sigma,
    sleep_variance,
    sleep_min,
    sleep_max,
    percent_1sigma,
    percent_2sigma,
    percent_3sigma
  )
)
part2_summary

```

```

##           Statistic      Value
## 1  Number of observations 58.000000
## 2                Mean    5.155172
## 3                Median    5.000000
## 4                Mode    4.000000
## 5 Sample standard deviation 1.694177
## 6                Variance    2.870236
## 7                Minimum    2.000000
## 8                Maximum    9.000000
## 9          Within 1 SD 62.068966
## 10         Within 2 SD 98.275862
## 11         Within 3 SD 100.000000

```

14. DISPLAY IMPORTANT RESULTS

```
cat("\nPART 2 RESULTS\n")
```

```
##
## PART 2 RESULTS

```

```
cat("Number of students:",
    length(sleep_hours), "\n")

```

```
## Number of students: 58

```

```
cat("Mean sleep duration:",
    round(mu, 4), "hours\n")

```

```
## Mean sleep duration: 5.1552 hours

```

```
cat("Standard deviation:",
    round(sigma, 4), "hours\n")

```

```
## Standard deviation: 1.6942 hours

```

```
cat("Within 1 SD:",  
    round(percent_1sigma, 2), "%\n")
```

```
## Within 1 SD: 62.07 %
```

```
cat("Within 2 SD:",  
    round(percent_2sigma, 2), "%\n")
```

```
## Within 2 SD: 98.28 %
```

```
cat("Within 3 SD:",  
    round(percent_3sigma, 2), "%\n")
```

```
## Within 3 SD: 100 %
```